

SAT UZBEKISTAN

Telegram @satuzbhunters

Module-1

1

Mark for Review



Lorenzo purchased a box of cereal and some strawberries at the grocery store. Lorenzo paid \$2 for the box of cereal and \$1.90 per pound for the strawberries. If Lorenzo paid a total of \$9.60 for the box of cereal and the strawberries, which of the following equations can be used to find p , the number of pounds of strawberries Lorenzo purchased? (Assume there is no sales tax.)

(A) $1.90p + 2 = 9.60$

(B) $1.90p - 2 = 9.60$

(C) $1.90 + 2p = 9.60$

(D) $1.90 - 2p = 9.60$

2

Mark for Review



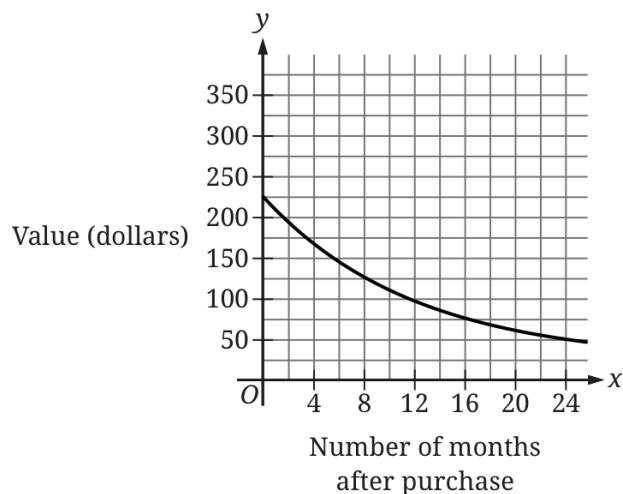
The function f is defined by $f(x) = 25x + 30$. What is the value of $f(x)$ when $x = 2$?

(A) 50

(B) 57

(C) 80

(D) 110



The graph shown gives the estimated value, in dollars, of a tablet as a function of the number of months since it was purchased. What is the best interpretation of the y -intercept of the graph in this context?

- (A) The estimated value of the tablet was \$225 when it was purchased.
- (B) The estimated value of the tablet 24 months after it was purchased was \$225.
- (C) The estimated value of the tablet had decreased by \$225 in the 24 months after it was purchased.
- (D) The estimated value of the tablet decreased by approximately 2.25% each year after it was purchased.

$$|p| + 61 = 65$$

Which value is a solution to the given equation?

- (A) $\frac{65}{61}$
- (B) 4
- (C) 126
- (D) 130

Triangles EFG and JKL are congruent, where E , F , and G correspond to J , K , and L , respectively. The measure of angle E is 45° and the measure of angle F is 20° . What is the measure of angle J ?

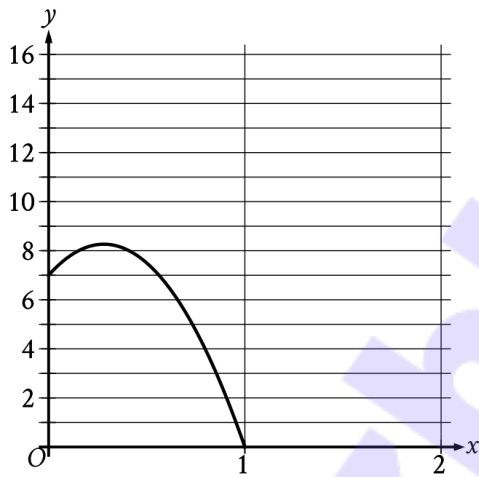
(A) 20°

(B) 45°

(C) 135°

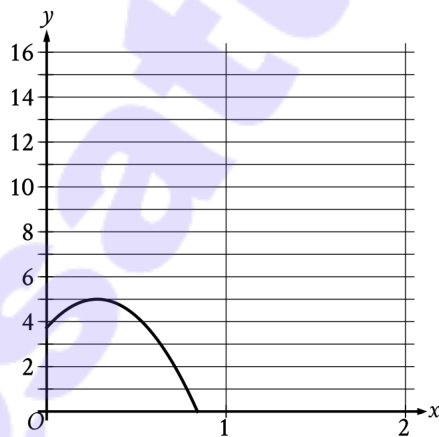
(D) 160°

During the first part of an experiment, a ball was launched from a 7-foot-tall platform. The graph shows the height y , in feet, of the ball x seconds after it was launched during the first part of the experiment.

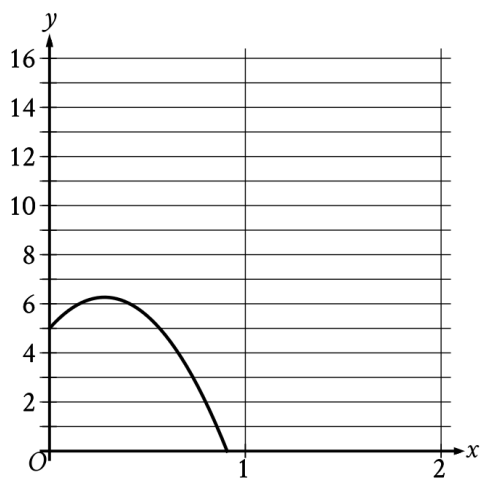


During the second part of the experiment, the ball was launched the same way, but from a platform that is 2 feet shorter than the first platform. Which of the following graphs could represent the height y , in feet, of the ball x seconds after it was launched during the second part of the experiment?

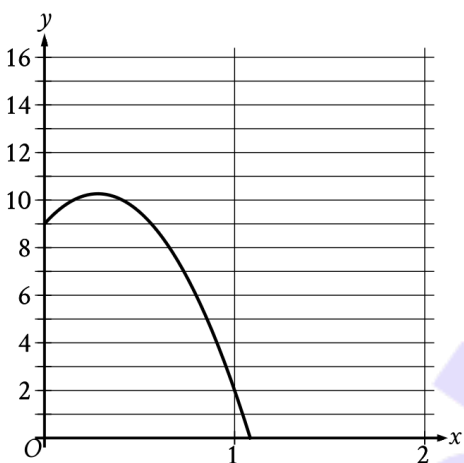
(A)



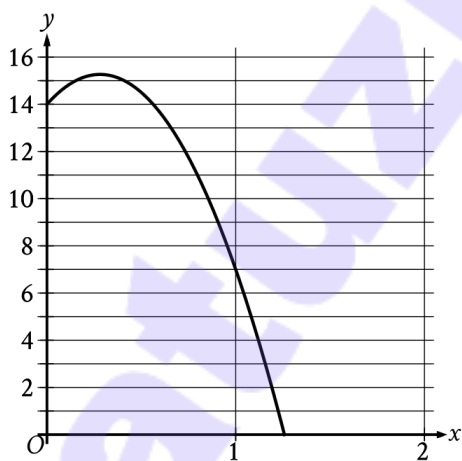
B



C



D



7

 Mark for Review

The ratio x to y is equivalent to the ratio 9 to 5. If the value of x is 162, what is the value of y ?

8

 Mark for Review

At a state fair, attendees can win tokens that are worth a different number of points depending on the shape. One attendee won S square tokens and C circle tokens worth a total of 1,120 points. The equation $80S + 90C = 1,120$ represents this situation. How many more points is a circle token worth than a square token?

(A) 950

(B) 90

(C) 80

(D) 10

9

 Mark for Review

$$\frac{(x+9)(x-9)}{x+9} = 7$$

What is the solution to the given equation?

(A) 7

(B) 9

(C) 16

(D) 63

10

Mark for Review



The equation $12t + b = c$ relates the variables t , b , and c . Which of the following correctly expresses the value of $c - b$ in terms of t ?

(A) $\frac{t}{12}$

(B) t

(C) $t + \frac{1}{12}$

(D) $12t$

11

Mark for Review

What is an x -coordinate of an x -intercept of the graph of $y = 3(x - 14)(x + 5)(x + 4)$ in the xy -plane?

12

Mark for Review

The function f is defined by $f(x) = 5\left(\frac{1}{4} - x\right)^2 + \frac{11}{4}$. What is the value of $f\left(\frac{1}{4}\right)$?

13

Mark for Review



A circle in the xy -plane has the equation $(x - 13)^2 + (y - k)^2 = 64$. Which of the following gives the center of the circle and its radius?

(A) The center is at $(13, k)$ and the radius is 8.

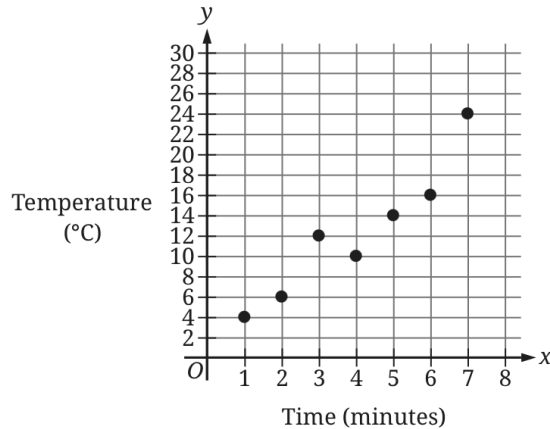
(B) The center is at $(k, 13)$ and the radius is 8.

(C) The center is at $(k, 13)$ and the radius is 64.

(D) The center is at $(13, k)$ and the radius is 64.

A circle has a radius of 2.1 inches. The area of the circle is $b\pi$ square inches, where b is a constant. What is the value of b ?

During a study, the temperature, in degrees Celsius ($^{\circ}\text{C}$), of the air in a chamber was recorded to the nearest integer at certain times. The scatterplot shows the recorded temperature y , in $^{\circ}\text{C}$, of the air in the chamber x minutes after the start of the study.



What was the average rate of change, in $^{\circ}\text{C}$ per minute, of the recorded temperature of the air in the chamber from $x = 5$ to $x = 7$?

Answer Preview:



$$3y = 4x + 17$$

$$-3y = 9x - 23$$

The solution to the given system of equations is (x, y) . What is the value of $39x$?

(A) -18

(B) -6

(C) 6

(D) 18

The function $f(t) = 40,000(2)^{\frac{t}{790}}$ gives the number of bacteria in a population t minutes after an initial observation. How much time, in minutes, does it take for the number of bacteria in the population to double?

(A) 2

(B) 790

(C) 1,580

(D) 40,000

If $5 - 7(2 - 4x) = 16 - 8(2 - 4x)$, what is the value of $2 - 4x$?

A scientist studying the life cycle of dragonflies counted the number of dragonflies in a certain habitat each day for 46 days. On February 15, there were 99 dragonflies in the habitat. The percent increase in the number of dragonflies in the habitat from January 1 to February 15 was 12.50%. How many dragonflies were in the habitat on January 1?

(A) 88

(B) 87

(C) 12

(D) 8

$$2x^2 - 8x - 7 = 0$$

One solution to the given equation can be written as $\frac{8-\sqrt{k}}{4}$, where k is a constant. What is the value of k ?

The function $f(x)$ is defined as 19 more than 4 times a number x . If $y = f(x)$ is graphed in the xy -plane, what is the best interpretation of the x -intercept?

- (A) When $f(x) = 0$, the number is $-\frac{19}{4}$.
- (B) When the number is 0, $f(x) = 19$.
- (C) The value of $f(x)$ increases by 1 for each increase of 4 in the value of the number.
- (D) For each increase of 1 in the value of the number, $f(x)$ increases by 4.

Which of the following expressions is equivalent to $(\sin 24^\circ)(\cos 66^\circ) + (\cos 24^\circ)(\sin 66^\circ)$?

- (A) $2(\cos 66^\circ)(\sin 24^\circ)$
- (B) $2(\cos 66^\circ) + 2(\cos 24^\circ)$
- (C) $(\cos 66^\circ)^2 + (\cos 24^\circ)^2$
- (D) $(\cos 66^\circ)^2 + (\sin 24^\circ)^2$

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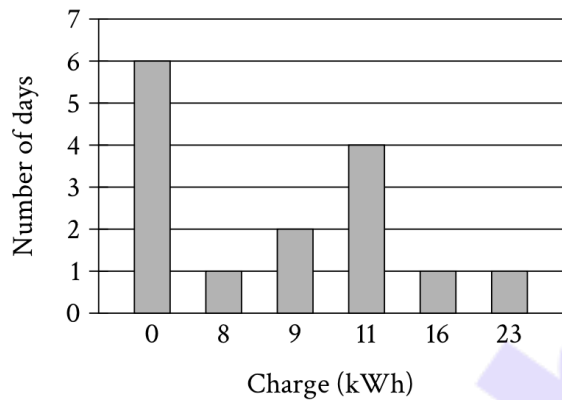
Module-2

1

Mark for Review



The bar graph summarizes the charge, in kilowatt-hours (kWh), a battery received each day for 15 days.



For how many of these 15 days did the battery receive a charge of 0 kWh?

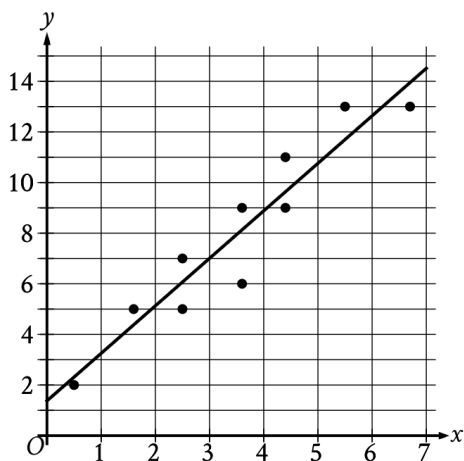
(A) 0

(B) 1

(C) 4

(D) 6

In the given scatterplot, a line of best fit for the data is shown.



Which of the following is closest to the slope of the line of best fit shown?

(A) 0

(B) $\frac{1}{2}$

(C) 1

(D) 2

In a sample, 80% of the items are faulty. There are 88 faulty items in the sample. How many total items are in the sample?

x	$g(x)$
-1	25
0	1
1	$\frac{1}{25}$
2	$\frac{1}{625}$

For the exponential function g , the table shows four values of x and their corresponding values of $g(x)$. Which equation defines g ?

(A) $g(x) = -25^x$

(B) $g(x) = -\left(\frac{1}{25}\right)^x$

(C) $g(x) = 25^x$

(D) $g(x) = \left(\frac{1}{25}\right)^x$

If $8x = 6$, what is the value of $72x$?

(A) 3

(B) 15

(C) 54

(D) 57

Which expression is equivalent to $\frac{h^{15}q^7}{h^5q^{21}}$, where $h > 0$ and $q > 0$?

(A) $\frac{h^{10}}{q^{14}}$

(B) $\frac{h^3}{q^3}$

(C) $h^{10}q^{14}$

(D) h^3q^3

A company fills boxes with approximately 23 pounds of oranges. To test the accuracy of the filling process, 344 boxes of oranges were selected at random and weighed. Based on the sample, it is estimated that the average weight of all boxes of oranges filled by the company in an 8-hour period is 23.1 pounds, with an associated margin of error of 0.19 pounds. Which of the following is the best interpretation of this estimate?

(A) Plausible values for the average weight of all boxes of oranges filled by the company are between 22.91 pounds and 23.29 pounds.

(B) Plausible values for the average weight of all boxes of oranges filled by the company are less than 22.91 pounds or greater than 23.29 pounds.

(C) The average weight of all boxes of oranges filled by the company is less than 23.01 pounds.

(D) The average weight of all boxes of oranges filled by the company is greater than 23.01 pounds.

8

Mark for Review

$$\frac{1}{3}(x+6) - \frac{1}{2}(x+6) = -8$$

What value of x is the solution to the given equation?

9

Mark for Review



$$f(x) = x^2 + bx$$

$$g(x) = 9x^2 - 27x$$

Functions f and g are given, and in function f , b is a constant. If

$f(x) \cdot g(x) = 9x^4 - 26x^3 - 3x^2$, what is the value of b ?

(A) -26

(B) $-\frac{26}{9}$

(C) $\frac{1}{9}$

(D) 9

10

Mark for Review



In the xy -plane, line s passes through the point $(0, 0)$ and is parallel to the line represented by the equation $y = 18x + 2$. If line s also passes through the point $(4, d)$, what is the value of d ?

(A) 2

(B) 18

(C) 72

(D) 74

11

Mark for Review



$$f(x) = (x - 44)(x - 46)$$

The function f is defined by the given equation. For what value of x does $f(x)$ reach its minimum?

(A) 46

(B) 45

(C) 44

(D) -1

In triangle XYZ , angle Y is a right angle, point P lies on $\overline{XZ}$, and point Q lies on $\overline{YZ}$ such that $\overline{PQ}$ is parallel to $\overline{XY}$. If the measure of angle XZY is 63° , what is the measure, in degrees, of angle XPQ ?



In a set of four consecutive odd integers, where the integers are ordered from least to greatest, the first integer is represented by x . The product of 12 and the fourth odd integer is at most 26 less than the sum of the first and third odd integers. Which inequality represents this situation?

- (A) $12(x + 6) \leq x + (x + 4) - 26$
- (B) $12(x + 6) \geq 26 - (x + (x + 4))$
- (C) $12(x + 4) \leq x + (x + 3) - 26$
- (D) $12(x + 4) \geq 26 - (x + (x + 3))$

$$\begin{aligned} \frac{2}{5}x + \frac{7}{5}y &= \frac{2}{7} \\ gx + ky &= \frac{5}{2} \end{aligned}$$

In the given system of equations, g and k are constants. The system has infinitely many solutions. What is the value of $\frac{g}{k}$?



x	y
$-2s$	24
$-s$	21
s	15

The table shows three values of x and their corresponding values of y , where s is a constant. There is a linear relationship between x and y . Which of the following equations represents this relationship?

- (A) $sx + 3y = 18s$
- (B) $3x + sy = 18s$
- (C) $3x + sy = 18$
- (D) $sx + 3y = 18$

$$y + k = x + 26$$

$$y - k = x^2 - 5x$$

In the given system of equations, k is a constant. The system has exactly one distinct real solution. What is the value of k ?



The function f is defined by $f(x) = \frac{1}{2}(x + 6)$. What is the value of $f(4)$?

(A) 20

(B) 12

(C) 10

(D) 5



The area of a rectangular banner is 2,661 square inches. The banner's length x , in inches, is 24 inches longer than its width, in inches. Which equation represents this situation?

(A) $0 = x^2 - 24x - 2,661$

(B) $0 = x^2 - 24x + 2,661$

(C) $0 = x^2 + 24x - 2,661$

(D) $0 = x^2 + 24x + 2,661$



A right circular cone has a volume of $71,148\pi$ cubic centimeters and the area of its base is $5,929\pi$ square centimeters. What is the slant height, in centimeters, of this cone?

(A) 12

(B) 36

(C) 77

(D) 85

$$7x + 6y = 5$$
$$28x + 24y = 20$$

For each real number r , which of the following points lies on the graph of each equation in the xy -plane for the given system?

(A) $(r, -\frac{6r}{7} + \frac{5}{7})$

(B) $(r, \frac{7r}{6} + \frac{5}{6})$

(C) $(\frac{r}{4} + 5, -\frac{r}{4} + 20)$

(D) $(-\frac{6r}{7} + \frac{5}{7}, r)$

The cost of renting a carpet cleaner is \$52 for the first day and \$26 for each additional day. Which of the following functions gives the cost $C(d)$, in dollars, of renting the carpet cleaner for d days, where d is a positive integer?

(A) $C(d) = 26d + 26$

(B) $C(d) = 26d + 52$

(C) $C(d) = 52d - 26$

(D) $C(d) = 52d + 78$

The speed of a vehicle is increasing at a rate of 7.3 meters per second squared. What is this rate, in **miles per minute squared**, rounded to the nearest tenth?
(Use 1 mile = 1,609 meters.)

(A) 0.3

(B) 16.3

(C) 195.8

(D) 220.4